

Advanced Programming – Exam 19 Jun 2026 - Part 1

Name:

Question 1 (1.2 points)

What is the difference between stack allocation and heap allocation in C++? Give one example of each.

Question 2 (1.2 points)

Explain the Rule of Three in C++. Why was it introduced?

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Question 3 (1.2 points)

What is runtime polymorphism and why is it useful? Write a small example using a base class `Shape` and derived classes `Circle` and `Square`.

Question 4 (1.2 points)

Explain template functions in C++. Write a templated function `maximum(x, y)`.

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Question 5 (1.2 points)

What is `std::optional`? When is it preferable w.r.to returning a pointer?

Question 6 (1.2 points)

Write a Python function that returns the unique elements of an input list.

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Question 7 (1.2 points)

Write a list comprehension that extracts all positive numbers from a list and squares them.

Question 8 (1.2 points)

Explain the purpose of `__iter__()` and `__next__()` in Python.

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Question 9 (1.2 points)

Using NumPy, compute the mean and standard deviation of an array.

Question 10 (1.2 points)

How can constructors be exposed to Python using pybind11?